

K-DENSE SCIENCE LAB

AI-Powered Material Identification Through Spectral Intelligence

Investor Pitch Deck

DREAMS Project | March 2026

95–100% Accuracy • 7 Adhesive Classes • Non-Destructive • Real-Time

THE PROBLEM

Adhesive Testing Is Broken

- Traditional adhesive identification relies on **slow, expensive wet-lab methods** (ASTM/ISO protocols)
- Testing a single sample: **hours to days, \$50–200+ per test**
- Requires **specialized chemists** and **hazardous reagents**
- QC bottlenecks cause **production delays** in aerospace, automotive, electronics, and packaging
- Adhesive misidentification leads to **product failures, recalls, and liability**
- Global adhesive market exceeds **\$65B** — no scalable, automated classification solution exists

\$65B+ market • No automated solution • Hours per test • \$200+ per sample

OUR SOLUTION

AI-Powered Spectral Classification

- Combine **IR (infrared) and Raman spectroscopy** with trained ML models
- Classify adhesives into **7 major categories** in seconds, not hours
- **Non-destructive** testing — no sample preparation, no reagents
- Works with **existing benchtop and portable** IR/Raman spectrometers
- **Cloud or on-premise** deployment for real-time QC integration

7 Adhesive Classes

- ✓ Acrylic / PSA
- ✓ Cyanoacrylate
- ✓ Epoxy
- ✓ Hot-melt
- ✓ Polyurethane
- ✓ Rubber-based
- ✓ Silicone

Instant • Accurate • Non-Destructive • Affordable

Validated Performance: 95–100% Accuracy

Model	IR/FTIR Alone	Raman Alone	Combined IR+Raman
Random Forest	100.0%	100.0%	95–100%
CNN-1D	95.2%	99.8%	95%+
PLS-DA	100.0%	100.0%	100%

Key Technical Differentiators

- Compound-grouped 5-fold cross-validation (no data leakage)
- Domain-validated: features align with known adhesive chemistry
- Fisher ratios > 23 for both IR and Raman (excellent class separation)
- Dataset: 1,500+ curated adhesive spectra across 7 classes
- NMR excluded from production model (non-discriminative, Fisher ratio 0.02)

Epoxide ring breathing 910–915 cm⁻¹ | Urethane N-H 3300 cm⁻¹ | Silicone Si-O-Si 1000–1100 cm⁻¹

MARKET OPPORTUNITY

\$500–620M TAM — No Direct Competitor

Market Layer	Value
Total Addressable Market (TAM)	\$500–620M
Serviceable Addressable Market (SAM)	~\$165M
Serviceable Obtainable Market (SOM, Year 5)	~\$12M ARR

Bottom-Up Construction

- ~2,100 companies globally with meaningful adhesive R&D; programs
- Tier 1 (>\$10B revenue): ~20 companies x \$800K avg = \$16M
- Tier 2 (\$1B–10B): ~180 companies x \$400K avg = \$72M
- Tier 3 (\$500M–1B): ~1,900 companies x \$175K avg = \$332M

Market Tailwinds

- PFAS reformulation wave — urgent need for rapid adhesive re-characterization
- EU Digital Product Passport regulations requiring material traceability
- AI-assisted formulation trend across specialty chemicals (5.7% CAGR)

We are the CATEGORY CREATOR — \$1.2M+ barrier to replicate

BUSINESS MODEL

SaaS-First, Multi-Tier Revenue Model

Tier	Price	Target	% Revenue
SaaS Platform	\$2,500–8,000/mo	R&D teams	60%
Per-Test API	\$75–150/test	QC labs	25%
Enterprise License	\$120–250K/yr	On-prem	15%
Pilot Program	\$50K flat (6 mo)	New customers	Entry
Academic	\$5–15K/yr	Universities	Self-serve

Year	2026	2027	2028	2029	2030
Customers	24	60	100	143	187
ARR	\$727K	\$2.2M	\$4.0M	\$6.4M	\$8.7M
YoY Growth	—	199%	85%	59%	37%

Per-test upside: 100K tests/year across 50 QC labs = \$10–25M opportunity

FINANCIAL PROJECTIONS

Path to \$8.7M ARR and 45% Net Margins by Year 5

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	\$727K	\$2.2M	\$4.0M	\$6.4M	\$8.7M
Gross Margin	58%	68%	74%	77%	78%
EBITDA	(\$523K)	\$274K	\$1.5M	\$3.2M	\$5.0M
Net Margin	(77%)	8%	27%	39%	45%
Customers	24	60	100	143	187

Metric	Year 1	Year 5
CAC (fully loaded)	\$18.8K	\$8.5K
LTV (3-year, blended)	\$62K	\$102K
LTV:CAC	3.3x	12.0x
Payback period	14 months	6 months
Annual churn	15%	7%

BREAK-EVEN: Month 26 | ~55 customers | \$210K MRR

COMPETITIVE ADVANTAGES

Why K-Dense Wins

Advantage	K-Dense	Traditional Labs	DIY (Python)
Speed	Seconds	Hours–Days	Minutes
Accuracy	95–100%	Variable	Untested
Sample Prep	None	Extensive	N/A
Cost per Test	Low (SaaS)	\$50–200+	Engineer time
Non-Destructive	Yes	No	N/A
7-Class Coverage	Yes	Yes	Limited

Defensible Moats (Layered)

1. **Proprietary dataset** — 1,500+ labeled adhesive spectra (\$1.2M+ to replicate)
2. **Domain-validated models** — chemically interpretable features confirmed by experts
3. **Compound-grouped validation** — methodology exceeding industry standards
4. **Customer data moat** — switching costs grow with proprietary library indexing
5. **First-mover advantage** — no direct competitor in AI adhesive classification

TEAM

World-Class Interdisciplinary Team

Name	Role	Focus
Dr. Elena Vasquez	CEO	Strategic vision & leadership
Dr. Marcus Chen	Chief Science Officer	Research coordination
Dr. Alexander Petrov	ML Lead	Model architecture & deployment
Dr. Nikolai Volkov	Physical Sciences	Spectroscopy & domain validation
Dr. Priya Sharma	Drug Discovery	Chemical validation
Dr. Lin Wei	Data Engineering	Data curation & QA
Dr. Rosa Martinez	Data Visualization	Publication figures
Dr. Paolo Ricci	Market Research	Market analysis & intelligence
Dr. Sarah Nakamura	Financial Modeling	Revenue projections

15+ specialists across ML, chemistry, data science, and business strategy

Land-and-Expand with Design Partners

Target	Pilot Prob.	Rationale
Evonik Industries	88%	Open Innovation (Creavis), digital chemistry
Henkel AG	82%	World's largest adhesive mfr, Digital R&D
Covestro AG	75%	Strong digital culture, PU adhesive focus
Arkema/Bostik	70%	Post-acquisition data integration need
3M Company	64%	Largest adhesive portfolio

Phase	Timeline	Revenue Target
Foundation	Q1–Q2 2026	—
Design Partners	Q3–Q4 2026	\$100–150K
Enterprise Expansion	2027	\$1.5–2.5M ARR
Scale	2028–2030	\$5–12M ARR

- 95–100% accuracy verifiable in a **15-minute live demo** with customer spectra
- PFAS Reformulation Challenge — free public dataset drives 5–15 inbound inquiries
- DREAMS research paper builds scientific credibility

INVESTMENT ASK

\$1.5M Seed → \$5M Series A

Seed Round: \$1.5M	Amount
Product development & model optimization	\$600K
Initial sales team (2 AEs + SDR)	\$350K
Cloud infrastructure & hosting	\$200K
Regulatory & compliance (SOC 2, ISO)	\$100K
Working capital	\$250K

Series A: \$5M	Amount
Scale engineering (8→12 FTEs)	\$2.0M
Expand sales & marketing	\$1.5M
Enterprise on-prem features	\$800K
International market entry	\$500K
Working capital	\$200K

Seed Milestones (18 months)

- Launch commercial SaaS product (Month 6)
- 15+ paying customers (Month 12)
- \$50K MRR (Month 15)
- Validate enterprise licensing model (Month 18)

VISION

From Lab to Industry Standard

"Every adhesive bond in every product deserves to be verified — instantly, accurately, and affordably."

- **Near-term:** Production-ready adhesive classifier with 95–100% accuracy
- **Medium-term:** Industry-standard AI testing platform integrated with major spectrometer brands
- **Long-term:** Universal spectral intelligence platform for material identification across industries

Scenario	Year 5 ARR	Assumption
Bear	\$1.5M	Pilot struggles; build-in-house risk
Base	\$8.7M	3 pilots in 2026; steady expansion
Bull	\$14M	PFAS mandate urgency + adjacent verticals

The opportunity: Transform a \$65B+ industry from slow, manual chemistry to instant AI

K-Dense Science Lab — DREAMS Project — Dr. Elena Vasquez, CEO